

PES UNIVERSITY

Department of Computer Science and Engineering

LAB 1 — REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

USE-CASE FLOW

Problem Statement #26 — Peer-to-Peer Carpooling & Expense Splitter

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Section:	C	Date:	27/08/2026

Use Case ID	UC-04	Primary Actor	Rider Commuter
Use Case	Request / Confirm Ride	Supporting Actor	Driver Host
Goal	Allow a rider to confirm a compatible available carpool seat.	Constraint	Maximum detour tolerance: 1.5 km

Main Success Scenario

1. The Rider Commuter enters the daily origin and destination.
2. The system searches for available Driver Host rides.
3. The system calculates route compatibility for each candidate ride.
4. The system excludes rides requiring a detour greater than 1.5 km.
5. The system displays compatible rides with departure time, available seats, and estimated fuel-cost share.
6. The Rider selects a suitable compatible ride.
7. The system verifies that at least one seat is currently available.
8. The Rider confirms the ride request.
9. The system reserves exactly one seat and records the Rider's participation.
10. The system calculates and displays the Rider's estimated fuel-cost share.
11. The system displays the confirmed ride details to the Rider.
12. The use case ends successfully.

Alternate Flows

A1 — Detour exceeds 1.5 km

- A11.** During compatibility calculation, the system determines that the required detour is greater than 1.5 km.
- A12.** The system marks the ride as incompatible and does not present it as a valid match.
- A13.** The Rider can select another compatible ride.

A2 — No seat available

- A21.** The Rider selects a ride that was previously available.
- A22.** The system detects that all seats have been taken.
- A23.** The system does not confirm the request and prompts the Rider to select another compatible ride.

Postcondition

A compatible carpool ride is confirmed, one seat is reserved for the Rider, the participation is recorded, and the estimated fuel-cost share is displayed.

Use-Case Traceability

FR-001 (1.5 km route compatibility) • FR-004 (seat request/confirmation) • FR-005 (fuel-cost share)